

CS3101: Programming and Data Structures I
Class Test Extra (Slot 2)

Starts: 29/11/2025 at 2 PM IST (Duration: 1 hour 45 mins)

Total marks: 25

Instructions

- All programs should be compatible with GNU C.
- Late submissions will attract marks deduction.
- You are advised to compile the code before submitting it to WeLearn. If a code does not compile at our end, marks will be deducted for the problem.
- Each program should follow a strict naming convention: **QNo.c** (e.g. Q1.c, Q2a.c etc.). Programs not adhering to the convention will not be evaluated.
- All codes (.c files) should be submitted in a single zipped folder to WeLearn.
- You should put appropriate comments in the code.

1. **(Marks: 10)** Write a program to implement a function `CanTransformK(s1, s2, k)` which checks if string *s1* can be transformed into another string *s2* in (minimum) *k* substitutions (*k* is 2). E.g., *King* can be transformed to *Rink* in *two* substitutions (*King* → **R**ing → *Rink*). You may call `CanTransformK()` from `main()` with arguments (*King, Rink, 2*) where the function should return 1, while for (*King, Pond, 2*) it should return 0 (as three substitutions are needed in this case). You are not allowed to use any built-in function for the same that can replicate `CanTransformK(s1, s2, k)`.
2. **(Marks: 15)** In a fun fair *ShukhSayar* there are three types of attractions (but with restrictions, to manage crowd): 1) Movie shows (max two can be watched per ticket): per show Rs. 300, 2) Rides (max two can be availed per ticket): per ride Rs. 150, 3) Games (max three can be played per ticket): per game Rs. 100. A person can buy only one ticket with her/his unique govt. ID, and it is not necessary for a person to choose from all the types of attractions (e.g., the person can only watch a movie or only have two rides, and nothing else, budget permitting).

Simulate an interactive interface in C which (after a nice welcome message) first displays the options and asks for the budget (in Rs.) from the user. The interface allows the user to choose items (e.g., 1 movie, 1 ride and 1 game is possible for budget Rs. 600, other combinations being possible) from the attractions (if budget permits; e.g., if the budget is less than Rs. 100, the user can't choose any), and after each valid choice, it prints the budget left, adhering to the restrictions (e.g. more than two movies can't be selected etc.), and adds the selected items to a **basket**. If the total item worth (of those currently in the basket + the current item chosen) for the person exceeds the remaining budget after a choice, you should issue a warning disallowing the choice, and that item should not be added to the basket, remaining budget unchanged. The user can also deselect a chosen item so that the basket and the remaining budget will be accordingly updated. You should ask the user after every choice if s/he wishes to continue.

Once the user is done with the selections, print the ticket, displaying the basket items and the residual budget, with a nice message. You may use only local variables; file handling is not necessary.